

EXPERIMENT - 13

MODELING CLOCK-DRIVEN CONCURRENT PROCESSES USING SC_THREAD AND WAIT() IN SYSTEMC

CODE:

```
#include <systemc.h>

SC_MODULE(ClockExample){
    sc_in<bool> clk;

    void process() {
        while (true){
            wait(clk.posedge_event());
            cout << "Time: "
                << sc_time_stamp()
                << " - Clock Triggered"
                << endl;
        }
    }

    SC_CTOR(ClockExample){
        SC_THREAD(process);
    }
};

int sc_main(int argc, char* argv[]){
    sc_clock clock("clock", 1, SC_NS);
    ClockExample module("module");
    module.clk(clock);
    sc_start(5, SC_NS);
}
```

```

return 0;

}

```

SIMULATION:

The screenshot shows the EDA Playground web interface. On the left, there's a sidebar with 'Languages & Libraries' (Testbench + Design, C++/SystemC, Libraries: SystemC 2.3.3) and 'Tools & Simulators' (Examples, 231). The main area has a code editor with the following C++ code:

```

1 #include <systemc.h>
2
3 SC_MODULE(ClockExample)
4 {
5     sc_in<bool> clk;
6
7     void process()
8     {
9         while (true)
10         {
11             wait(clk.posedge_event());
12
13             cout << "Time: "
14                  << sc_time_stamp()
15                  << " - Clock Triggered"
16                  << endl;
17         }
18     }
19
20     SC_CTOR(ClockExample)
21     {
22         SC_THREAD(process);
23     }
24 };
25
26 int sc_main(int argc, char* argv[])
27 {
28     sc_clock clock("clock", 1, SC_NS);
29     ClockExample module("module");
30 }

```

Below the code editor is a terminal window showing the compilation and simulation process:

```

[2026-08-03 09:02:55 UTC] g++ -o sim -lsystemc *.cpp && echo "Compile done. Starting run..." && ./sim
Compile done. Starting run...

SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
Copyright (c) 1996-2018 by all Contributors,
ALL RIGHTS RESERVED
Time: 0 s - Clock Triggered
Time: 1 ns - Clock Triggered
Time: 2 ns - Clock Triggered
Time: 3 ns - Clock Triggered
Time: 4 ns - Clock Triggered
Done

```

OUTPUT:

This screenshot shows a close-up of the terminal window from the previous image. It displays the same output, including the compilation command, SystemC version information, and the simulation results showing clock triggers at 0s, 1ns, 2ns, 3ns, and 4ns, followed by a 'Done' status.

```

[2026-08-03 09:02:55 UTC] g++ -o sim -lsystemc *.cpp && echo "Compile done. Starting run..." && ./sim
Compile done. Starting run...

SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
Copyright (c) 1996-2018 by all Contributors,
ALL RIGHTS RESERVED
Time: 0 s - Clock Triggered
Time: 1 ns - Clock Triggered
Time: 2 ns - Clock Triggered
Time: 3 ns - Clock Triggered
Time: 4 ns - Clock Triggered
Done

```